

CSCE 5380 DataMining

Project 01 Milestone 2

A. TASK 04 :

We have injected the noise into the dataset and repeated the pre processing steps again on the noisy data set.

Now we repeat the process of Cross validation on the three models again with the noisy data and got accuracies any repeated the ANOVIA + Turkey test.

The accuracies are:

DT	PART	JRip
0.6607996	0.6153074	0.6581003

The ANOVIA TABLE:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
Model	2	0.03908	0.019540	7.148	0.00134	**
Residuals	87	0.23782	0.002734			

Significance. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ANOVA Test Result: significant difference ($p < 0.05$). So we do a turkey test. Tukey multiple comparisons of means at 95% family-wise confidence level

Fit: aov(formula = Accuracy ~ Model, data = acc_df_n)

	diff	lwr	upr	P adj
JRip-DecisionTree	-0.002699288	-0.03488883	0.02949025	0.9782057
PART-DecisionTree	-0.045492180	-0.07768172	-0.01330264	0.0031914
PART-JRip	-0.042792893	-0.07498243	-0.01060335	0.0059106

The mean accuracy of the models at 10% Noise :

Model <fct>	mean_accuracy <dbl>
DecisionTree	0.661
JRip	0.658
PART	0.615

So we can say that decision tree is more robust to noise.

B. Observations on these Experiments:

Now we are going to evaluate these 3 classifier based on mentioned 4 features

1. Feature 1: Rule Perspicacity:

Here it means how short and intrerpetable the rules are to understand. Decision Tree has more rules and is not easily interpretable so it would be rank 3, compared to to DT, Part has less rules (52 rules) and moderately interpretable with some redundancies so, I wold rank it 2 and Ripper has 4 rules and can be interpreted easily.

Here RIPPER(1)> PART(2)> DT(3)

2. Feature 2: Mean Accuracy:

From the Milestone project we got the mean accuracy of the models to be

- i. JRip = 0.744
- ii. DecisionTree = 0.723
- iii. PART = 0.705

From we can rank JRip as 1 DT as 2 and PART as 3

So, Here RIPPER(1)> DT(2)> PART(3)

3. Feature 3: Stability:

It about the spread accuracy across the folds from the Milestone we can see all the 30 values of each classifier model and decide the ranking based on the variance :

Measure	Decision Tree	PART	Ripper
Mean accuracy	0.723333337	0.705238087	0.744285717
Varaiance accuracy	0.001780202	0.002568379	0.002833919

From we can say that DT is more stable as it has less variance across the folds.

ranking is DT(1)> PART(2)> RIPPER(3)

4. Feature 4:Robust To Noise:

mean accuracies _noise:

- 1 DecisionTree 0.661
- 2 JRip 0.658
- 3 PART 0.615

accuracy drops: mean_acc-Mean_acc_noise

The least drop in accuracy means more robust to noise.

Model Drop

DT 0.0623

JRip 0.0863

PART 0.0902

ranking DT has less drop meaning more robust to noise

ranking: DT(1)>RIP(2)>Part(3)

Now Lets create a table

Model	Rules Perspicacity	Mean Accuracy	Stability	Robust to noise	average
DT	3	2	1	1	1.75
PART	2	3	2	3	2.5
RIPPER	1	1	3	2	1.75

So we can say that both DT and RIPPER has same priority we can use DT if we more stable and robust model to noise , Or we can use JRIP we want the interpretable and high accuracy of the model.

For data mining we need more stable model than accuracy, so we can say that DT>RIPPER>PART.